

SPATIAL INTEROPERABILITY DEMONSTRATION PROJECT (SIDP)

FINAL PARTICIPANTS REPORT

- 1.0
- 24 May 2005



SPATIAL INTEROPERABILITY DEMONSTRATION PROJECT (SIDP)

FINAL PARTICIPANTS REPORT

- 1.0
- 24 May 2005

Sinclair Knight Merz
ABN 37 001 024 095
100 Christie Street
PO Box 164
St Leonards NSW
Australia 1590
Tel: +61 2 9928 2100
Fax: +61 2 9928 2500
Web: www.skmconsulting.com

COPYRIGHT: The concepts and information contained in this document are the property of Sinclair Knight Merz Pty Ltd. Use or copying of this document in whole or in part without the written permission of Sinclair Knight Merz constitutes an infringement of copyright.



Contents

Summary	1
Acknowledgements	3
1. Introduction	6
2. SIDP Roadshow	7
2.1 Overview	7
2.2 Evaluation	8
2.3 Issues raised	9
2.4 Recommendations	10
3. Critical Success Factors	11
4. Lessons Learned and Next Steps	19
4.1 Lessons learned	19
4.2 Next steps	20
Appendix A Detailed Survey Results	21
A.1 Summary of responses	21
A.2 Other Comments	28
Appendix B ‘Burning Questions’ raised in seminars	32
B.1 Hobart	32
B.2 Melbourne	32
B.3 Sydney	32
B.4 Canberra	33
B.5 Brisbane	33
B.6 Adelaide	34
B.7 Perth	34



Document history and status

Revision	Date issued	Reviewed by	Approved by	Date approved	Revision type
1.0	24 May 2005	MvdV	n/a		Final Report

Distribution of copies

Revision	Copy no	Quantity	Issued to
1.0			ASIBA, SIDP Steering Committee, Participants

Printed:	24 May 2005
Last saved:	24 May 2005 12:27 PM
File name:	I:\ENVR\Projects\EN01726\Project Management\Project Reports\Final SC progress report\SIDP_Final_Report_1.0.doc
Author:	Maurits van der Vlugt
Project manager:	Maurits van der Vlugt
Name of organisation:	ASIBA
Name of project:	SIDP
Name of document:	Final Participants Report
Document version:	1.0
Project number:	EN01726



Summary

The Spatial Interoperability Demonstration Project (SIDP) ran from February 2004 until May 2005, delivering live demonstrations, training and educational materials that show the relevance, reality and robustness of spatial information interoperability to users, implementers and service providers in the fields of Emergency Management, Insurance and Utilities. The SIDP approach has proven to be very powerful, compelling and convincing.

The main achievements of the project are:

- The successful delivery of a series of demonstrations and seminars throughout Australia, attracting over 500 attendees, and receiving extremely positive feedback from the attendees
- Delivery of a 'Resource Kit' to support capacity building in hardcopy, on CD and online.
- Continuing online presence of technical resources and demonstrations for the next 12 months
- Successful cooperation between 30 organisations from the public and private sectors in delivering the project
- Identification of the most important 'burning issues' that need further attention

Project deliverables are publicly available and include:

- User requirements for interoperable web services in Emergency Management
- Demonstration scenarios and storyboards
- 'Blueprint' for implementing interoperable web services, in the form of three architecture documents
- Live, online demonstrations
- SIDP resource kit
- Workshops and seminars conducted in all capital cities

Specific outcomes for the spatial information industry are the development of market opportunities, supported by a wide range of interoperable web services that have been stood up in SIDP, and the increased capacity to implement interoperable solutions.

The performance of the project measures very well against the planned deliverables and the predefined critical success factors. As illustrated in section 3, SIDP has met, and in many cases overachieved, its targets. Of the surveyed workshop participants, 99.7% found the workshops useful, more than 80% found the scenarios and seminar relevant or better and the proportion of participants who indicated a "good" or "very good" knowledge level of Interoperable Web Services



increased dramatically by attending the seminar (from 28.8% before, to 83.1% after). Ten different SIDP data providers hosted hundreds of spatial data layers as Interoperable Web Services, more than 30 of which were deployed in the demonstrators.

SIDP has been particularly successful in delivering convincing, real-world demonstrations and accompanying these with good and attractive educational materials. Feed-back from participants was overwhelmingly positive, and included comments like:

- “Very professional presentation of a complex subject. Excellent material with substance. Not just a showcase - a real kick starter...”,
- “Well done - at last a practical, relevant proof of concept.”, and
- “Very interesting. I knew nothing about this subject and now I know quite a lot and have material to learn further.”

The success and popularity of the SIDP Resource Kits is further illustrated by the fact that one participating company is already printing 400 extra copies of the Resource CD for distribution among its customers, and ANZLIC has ordered 100 extra copies of the full Resource Kit.

The outcomes of SIDP are also acknowledged internationally, where there is a high level of interest to repeat the SIDP roadshow in New Zealand, and the Open Geospatial Consortium (OGC) has created a Risk and Crisis Management Working Group (RCM WG) which will build upon the SIDP scenarios.

Despite its success, a few points for improvement have been identified in section 4.1. The most notable is that SIDP has had only limited success in reaching out to the non-traditional markets of utilities and insurance. This is currently being addressed.

SIDP has identified a number of issues as the most important to be tackled in any follow-up activities. These are both technical (e.g. security, digital rights management, catalogues) and institutional (e.g. privacy, data quality, IP, costing and charging) in nature.

SIDP as a project will finalise the last workshop in Darwin (8 July) and continue to have an online presence until June 2006.



Acknowledgements

ASIBA, as principal sponsor would like to thank the following individuals and organisations for their contribution to making this project a success.

Project Sponsors

- AusIndustry - Innovation Access Program (IAccP)
- Open Geospatial Consortium Australasia (OGC-A)

Project Manager

- Mr. Maurits van der Vlugt, Senior Geo-IT Consultant, Sinclair Knight Merz

SIDP Steering Committee

- Mr. Paul Kelly (Chair) Consultant, Spatial Strategies
- Mr. David Hocking CEO Australian Spatial Information Business Associations (ASIBA)
- Mr. Rob Starling Open Geospatial Consortium (Australasia) OGC-A
- Mr. Jack de Lange Chief Operating Officer (ASIBA)
- Dr. James Moody Director, Divisional Business Strategy, CSIRO Land and Water
- Mr. Morris Bradley Director Knowledge Management & Business for Emergency Management Australia (EMA)
- Mr. Paul Trezise Chief Information Officer of Geoscience Australia
- Mr. Chris Gentle Consultant, Aurora Australis
- Mr. Phil Beach Manager – Shared Land Services, WA - Department of Land Information

Industry Advisory Committee

- Mr David Thomas Bureau of Meteorology
- Mr Dragi Neskovski Munich Reinsurance
- Mr Morrie Bradley Emergency Management Australia

Technical Advisory Committee

- Dr Andrew Bond DSTC (Distributed Computing)
- Dr Stephen Blake ANZLIC
- Ms. Jenny Bone ANZLIC
- Mr Aaron Sedgmen Corporate Information Management and Access, Geoscience Australia
- Dr Carl Reed OGC
- Dr Stephan Winter University of Melbourne



- Dr Robert Dahni Bureau of Meteorology

Private Sector Participants

- Compusult
- ER-Mapper
- ESRI Australia
- Homer Systems
- Kay consulting
- MapInfo Australia
- mapShed
- MapWerks consortium (incl. Lisasoft & Terrapages)
- NGIS (incl. Ionic software)
- Omnilink
- PSMA
- SKM
- Social Change Online
- WebMap

Public Sector Participants

- ACT Government and ICSM
- Bureau of Meteorology
- Commonwealth Scientific and Industrial Research Organisation (CSIRO)
- Cooperative Research Center for Spatial Information (CRC-SI)
- Curtin University Department of Spatial Sciences
- Geoscience Australia
- NSW Dept of Lands
- NSW Dept of Infrastructure, Planning and Natural Resources
- VIC Country Fire Authority (CFA)
- WA Department of Land Information (DLI)

Partner Projects

- CRC-SI Emergency Management Demonstrator Project (EMDP)
 - Ms. Kylie Armstrong
 - Mr. Mark Carniello
- Minerals Council of Australia, Solid Earth and Environment Grid project (MCA SEEGRid)



- Mr. Rob Atkinson
- Dr. Simon Cox
- Mr. Andrew Dent
- Mr. Stuart Girvan
- Dr. Lesley Weyborn
- Dr. Robert Woodcock



1. Introduction

This document constitutes the final report by the SIDP project managers to the project participants. It addresses the outcomes of the project, measured against the project and communications plans, and specifically against the identified critical success factors. Furthermore, it examines the lessons learned, and proposes next steps.



2. SIDP Roadshow

2.1 Overview

The SIDP roadshow (workshop series-2) was designed to address to distinct audiences. A first ‘executive briefing’ part explaining the “what” and “why” of interoperability, including the demonstrations. Followed by a seminar, a more technical “how” session, aimed at the implementation level. It turns out however, that over 70% of attendees attended both sessions.

The roadshow attracted well over 500 attendees from a wide range of sectors, as illustrated in the tables below. Best attendance was achieved in WA, but even the Tasmanian sessions were well attended considering the size of that jurisdiction.

■ **Table 1 Overview of workshop and seminar attendance**

State/Territory	Total Registrations	Attendance	Comments
ACT	113	76	Estimate, no actual attendance recorded at venue
NSW	103	74	
QLD	106	90	
SA	95	66	
TAS	61	48	
VIC	90	65	
WA	133	101	
TOTAL	701	520	

Note that due to initial lack of interest, the NT/Darwin sessions have been rescheduled to 8 July, to coincide with the North Australian Remote Sensing and GIS forum (NARGIS).

■ **Table 2 Attendee's sector(s) - from post workshop survey**

Sector	Number of responses	Response percent
State/Territory Government	71	24%
Spatial Information Industry	67	22%
Federal Government	47	16%
Emergency Services	46	15%
Other (please specify) ¹	44	15%
Environmental Management / Monitoring	28	9%
Local Government	25	8%
Utilities	20	7%

¹ Mainly consulting, mapping, and academia



Sector	Number of responses	Response percent
Transport	16	5%
Resource Inventory / Mining	14	5%
Law and Order	13	4%
Land Titling / Administration	12	4%
Urban Development / Planning	12	4%
Land Development	11	4%
Asset / Facilities Management	10	3%
Education	8	3%
Insurance	7	2%
Agriculture	6	2%
Civil / Building Construction	6	2%
Medicine	5	2%
Defence	4	1%
Indigenous Lands	3	1%
Social Services	3	1%
Telecommunications	2	1%
Tourism	2	1%
Logistics	1	0%
Marketing / Market Research	1	0%
Retail	1	0%
Culture & the Arts	0	0%

We have had limited success in getting the attention from the utilities and insurance industries, despite actively disseminating information on our project through dedicated channels: GITA (Geospatial Information Technology Association) and ICA (Insurance Council of Australia) respectively.

2.2 Evaluation

The feedback from attendees was overwhelmingly positive, as illustrated in detail in the detailed post workshop survey results listed in Appendix A. A near perfect 99.7% rated the workshop as worthwhile, with over 95% indicating that it met or exceeded their expectations. Most comments that were made were very positive, some containing constructive criticism. Some examples are listed below (a full listing is available in Appendix A).

■ Table 3 Selected workshop comments

State	"Other Comments"
ACT	Discoverability and fitness of use of data and services were not properly addressed.
ACT	Well done. Where to next?
Qld	Very professional presentation of a complex subject. Excellent material with substance. Not just a



State	"Other Comments"
	showcase - a real kick starter Now we need to focus on the complex institutional arrangements and build momentum for change.
Qld	Well done - at last a practical, relevant proof of concept.
SA	Good, reinforced that interoperability capability is real, robust, proven. It has been in use/published for 2+ years. What SIDP showed is not new, but it did increase awareness to non-gis people.
SA	Good - Understandable by someone with minimum exposure to GIS/Mapping Skills.
SA	Excellent to see that the profile of spatial information is being raised and discussed with a business focus. Hopefully these types of forums will allow a structured growth of spatial information.
Tas	Workshop was a good starting point however further forums in this area need to be held to facilitate the adoption of standards and data sharing in Tasmania. A regional sub-group of ASIBA forming was mentioned but this excludes the major stakeholders such as governments, so this needs to be addressed also. Government hold most of the datasets.
Tas	Demonstration/scenarios were very professionally presented as was facilitation. Well done.
Vic	Excellent presentation and content of training materials
Vic	I didn't enjoy the 'business' sessions where the audience was allowed to discuss/whine about costs and licensing. SIDP was the topic.
Vic	Very interesting. I knew nothing about this subject and now i know quite a lot and have material to learn further.
WA	Time well spent in obtaining information about SIDP. A very useful presentation and workshop.

2.3 Issues raised

At the beginning of each Seminar session, participants were asked by the facilitator what questions they wanted answered during the session: called 'Burning Questions'. The facilitator ensured that these questions were answered by either the technical or business-related sessions in the Seminar.

The Burning Questions asked during each Seminar are shown in Appendix B. In general, the questions asked reflected the confidence/competence of the participants in Spatial Information Management Systems in general and Spatial Interoperability in particular. The Burning Questions also reflected the particular initiatives and historical context within each jurisdiction.

Despite the potential for wide variation in Burning Questions for the reasons outlined above, a number of common themes emerged. These were:

- Where do you find web services?
- How is the data quality, currency and integrity of web services known to users?
- How are charging models developed and applied (both for consumers and producers of services)?
- How is security managed?
- How much does this all cost?



- How is risk management, contingency planning and disaster recovery managed?
- How is liability, copyright and intellectual property managed?
- What are the differences between standards, how are these updated and how best to start implementation?

This suggests that follow-up projects for ASIBA in Spatial Interoperability should focus on answering the above questions – in particular from a business management perspective.

2.4 Recommendations

The Roadshow component of the SIDP was a major success. The feedback from participants (incl. on the Resource Kit) was extremely favourable. The Executive Briefings and Seminars were well received. The Evaluation Sheets completed by participants (to be analysed by the Project Manager) will provide further details.

In the interests of improving future projects of this kind, a number of recommendations were made by Kay Consulting, who were responsible for delivering the SIDP training and education component:

- 1) Education and training components of projects should not be viewed as an element that is ‘tacked on the end’ of project; or if they are, then sufficient time is allocated for their completion.
- 2) The inter-dependencies between education and training components and other project elements are more effectively project managed and the appropriate time allocated.
- 3) Seminar venues and locations be chosen to ensure maximum participation of target audiences and to better meet the needs of the seminar presenters.
- 4) Follow-up projects of the SIDP to focus on:
 - Technical implementation issues – especially security, digital rights management and catalogues
 - Managerial issues – especially security, IP, cost, charging, new business models, information access and liability



3. Critical Success Factors

This section presents the set of success criteria for SIDP. It is based on the Goals, Outcomes and Tasks table as defined in the SIDP Detailed Project Plan (Section 2.4). Added to this are the bolded columns in the table below. Listed for each outcome are:

- The Critical Success Factor: the measurable entity that we use to define a successful outcome
- The Target: the value of the success factor that SIDP considers to indicate a satisfactory, good or exceptional outcome respectively.
- How to Measure: by what method SIDP will measure the factors, and
- When to Measure: when the measurement will take place, and when SIDP will report the outcomes to the Steering Committee
- The metric outcome and achievement level

A number of success factors (12-14) refer to 'awareness raising', and have been measured by conducting an industry survey before and after the demonstrations.

Success factors 13 and 15 refer to take-up in the period after SIDP completion, and will need to be reviewed at the end of 2005.

Table 4 shows that the SIDP passed all identified critical success factors, the majority of them as 'good' or 'excellent'.

■ **Table 4: SIDP Success Factors**

Goal	Outcome	Task	Critical Success Factor	Target S = Satisfactory G = Good E = Exceptional	How to Measure	Metric Result	Achievement level
The co-ordination of Australia's spatial data providers, users and service providers to confirm common interoperability standards and systems	Comprehensive list Technical issues and standards for the SIDP;	Host workshops with the Information owners and the implementers	1) Number of attendees at workshops and seminars in Phase 4	S – 150 G – 200 E – 250	List of attendees	520 Participants in series-2 workshops & seminars	Excellent
			2) Number of committed participants with the SIDP	S – 8 G – 10 E – 15	Signed participants with in kind contributions	■ 15 Industry participants² ■ NSW DIPNR ■ NSW Dept of Lands ■ VIC CFA ■ Bureau of Meteorology ■ Geoscience Australia ■ WA DLI	Excellent

² ESRI Australia, ER-Mapper, NGIS, MapWerks, MapInfo Australia, Kay consulting, mapShed, Homer Systems, Omnilink, SKM, WebMap, PSMA, Compusult, Social Change Online, Ionic.
SINCLAIR KNIGHT MERZ

Goal	Outcome	Task	Critical Success Factor	Target S = Satisfactory G = Good E = Exceptional	How to Measure	Metric Result	Achievement level
	Final implementation plans to support the SIDP real life scenarios across the 3 sectors	Host workshops with customer groups to capture requirements	3) Number of published scenarios	3 Scenarios published	Acceptance of the scenarios by the Steering Committee	2 Scenarios published by merging scenarios #1 (Bushfire management) and #2 (Utility notification)	Pass
			4) The number of attendees at customer workshops who provide input into clearly defined implementable scenarios	S – 150 G – 175 E – 200	Attendance lists	197 participants in series-1 workshops (July – August 2004)	Excellent
	Definition of the constraints and opportunities in relation to technology supply during SIDP	Functional Requirements specification	5) Acceptance by the Steering Committee and the Technical Advisory Committee	Acceptance	Review by Technical Advisory Committee	Functional Requirements documented in series-1 workshop outcomes report and Scenario storyboards (published on http://sidp.com.au)	Pass

Goal	Outcome	Task	Critical Success Factor	Target S = Satisfactory G = Good E = Exceptional	How to Measure	Metric Result	Achievement level
	Maximise linkages to complementary concurrent projects	Specific workshops with the CRC EMDP and MCA-SeeGrid to define the co-operation, shared tasks and shared information	6) Number of, and attendance at, joint workshops	WS Att S – 2 G – 4 E – 6 60 150 300	Attendance lists	- SIDP PM attendance at two SeeGrid workshops - CRC-SI EMDP co-presentation at five SIDP series-2 workshops (400³ attendees)	Good – Excellent
The comprehensive discovery of the existence, source and quality of the spatial data infrastructure across Australia	Web enabled catalogue of configured data sources to support the Demonstration across all 3 scenarios + linkage projects	Contact each participating organisation as well as other key information owners and configure the data sources to support SIDP	7) Number of participating Information Owners	S – 5 G – 8 E – 12	Number of signed information owners	Private sector: - PSMA - MapInfo - SKM - WebMap - ER-Mapper Public sector: - NSW DIPNR - NSW Dept of Lands - WA DLI - Bureau of Meteorology - Geoscience	Good - Excellent

³ Approximate, exact number to be confirmed
SINCLAIR KNIGHT MERZ

Goal	Outcome	Task	Critical Success Factor	Target S = Satisfactory G = Good E = Exceptional	How to Measure	Metric Result	Achievement level
	To provide clarity on the ongoing access to spatial information, post SIDP.	Engage each provider through one on one discussions to define the terms, cost and availability of data for the scenario development and on-going basis	8) Number of data supply agreements for demonstration and beyond	S – 5 G – 8 E – 12	Number of signed agreements	Australia See above	Good-Excellent
Demonstrate the technologies and the benefits available currently for implementing interoperable spatial data and information services	Technology platform to demonstrate the scenarios for each industry	Construct Open GIS Demonstrations	9) The number of data layers from different host organisations deployed in the different scenarios.	S – 20 G – 30 E – 40	Review demonstration applications and/or count published data sets in registry	Hundreds of data layers served, Bureau of Meteorology alone serves 50 layers Deployed data layers: ■ Scenario 1: 8-10 ■ Scenario 2: 17-20	Good
	Refined user interface and data configuration to maximise value and user acceptance	Obtain user feedback via user acceptance testing	10) Customer Feedback on relevance of scenarios and technology to their operations	Proportion of responses "Relevant or better" S – 50% G – 65% E – 80%+	Feedback forms completed at conclusion of demonstration	Scenarios (73.7%) ■ Relevant: 46.3 % ■ Very relevant 27.4 % Technology (90.8%) ■ Relevant: 53% ■ Very relevant: 37.8% Seminars (86.9%)	Excellent

SINCLAIR KNIGHT MERZ

Goal	Outcome	Task	Critical Success Factor	Target S = Satisfactory G = Good E = Exceptional	How to Measure	Metric Result	Achievement level
	Demonstrated integration with CRC-EMDP and MCA-SeeGRID	Network and interface with different linkage projects identified under section 2.4 of the detailed project plan	11) The number of linkage projects actively involved with or influenced by SIDP	S – 4 G – 6 E – 8	Review by Steering Committee Chairman	- Relevant: 28.6% - Very relevant: 58.3% Integrated CRC-SI EMDP and SeeGRID in workshop series-2. Technical Architecture developed with SeeGRID Demonstrators showed use of WA-SLIP/DLI services Notional Architecture and Architecture Executive Summary deliverables shared with ANZLIC/ASDI	Satisfactory
Creation of awareness about the benefits and feasibility of spatial information and its interoperability in the three plus one customer industry sectors	Networked and interfaced with different linkage projects identified under section 2.4 of the detailed project plan	Workshops with the customer	12) Increased awareness in the customer industry sectors at the operations management level	S – 10% increase G – 20% increase E – 30% increase	Measured by a before and after industry survey	“How would you rate your knowledge of SIDP and interoperable web services?” % respondents who answered “Good” or “Very good”: - Before: 28.8 % - After: 83.1 %	Excellent

Goal	Outcome	Task	Critical Success Factor	Target S = Satisfactory G = Good E = Exceptional	How to Measure	Metric Result	Achievement level
Promoted the concept of interoperability to the benefit of other industries and thereby maximise the use and potential of geospatial information and interoperable technologies	Conducted a series of workshops Communicated the outcomes of the project to stakeholders and the general public	Communications strategy involving, formal launches in each State press releases, television and workshops	13) Amount of "market pull" (new types of users of geospatial information that emerge during 2005)	S – 10% increase G – 20% increase E – 30% increase	Measured by a before and after industry survey	To be measured end of 2005	n/a
Created and shared the "know how" of implementing interoperable geospatial solutions	Developed and distributed training and communication materials	Creation of the SIDP website populated with manuals, cookbooks, user scenarios and project highlights	14) The materials are of sufficient quality for new entrants from the SI industry being able to implement interoperable web services	Proportion of responses "Sufficient Quality" S – 50% G – 65% E – 80%+	Feedback forms completed at conclusion of technical seminars	"Please rate the relevance/usefulness of the training kit for your purposes". % Of respondents who answered "Relevant" or "Very relevant": 76.2%	Good – Excellent
Serve as an ongoing industry focus for information, advice and services to facilitate the application and use	Ongoing interoperable environment that continues to expand	Negotiate arrangements with Information Owners to continue to operate the standards compliant portals	15) Interoperable standards are specified as a design requirement into Geospatial	S – 20% G – 45% E – 70%+	ANZLIC ASDI reporting	To be measured end of 2005	n/a

SINCLAIR KNIGHT MERZ

Goal	Outcome	Task	Critical Success Factor	Target S = Satisfactory G = Good E = Exceptional	How to Measure	Metric Result	Achievement level
of interoperable spatial data by and within Australian industry			portal implementations that commence after the completion of SIDP				



4. Lessons Learned and Next Steps

4.1 Lessons learned

This section lists a number of lessons learned from the SIDP. Some of these contain suggestions for improvement, and are identified with the blessings of 20-20 hindsight.

- 1) The SIDP approach of using relevant, real, robust and pervasive demonstrations have proven to be very powerful, compelling and convincing.
- 2) The spatial information industry's maturity has been demonstrated by the wide take-up of interoperable capabilities and the willingness of normally competing companies to interoperate within SIDP
- 3) Two of the most fundamental interoperability prerequisites are now pervasive: reliable network connections and availability of robust, off-the-shelf software components that implement interoperability standards.
- 4) SIDP has had limited success in reaching out to non-traditional markets. Notably the insurance and utilities industries. Further communication and outreach to these sectors may be called upon. In future projects, care should be taken to engage champions of these sectors early on, and more actively promote the project to the target sectors
- 5) The SIDP roadshows particular design implied specific venue requirements that allowed for large and highly visible projection capabilities for the demos, while at the same time enabling active audience participation. This was not always available, in part as a result of budget constraints. Some of the venues were either inconvenient for senior managers to get too, or were too small. At Brisbane, people had to be turned away as the venue could only accommodate limited numbers.
In future projects, venue requirements will need to be specified in detail early on, so appropriate venues and logistical arrangements can be organised. It will also be highly convenient to engage a single, national staging company to arrange the set-up, network and audio-visual arrangements at all venues.
- 6) Continuity of the project team is important. In SIDP this was most notably absent in the technical/architectural tasks, where notional, technical and implementation architectures were developed by three different organisations. Future projects of this size would benefit from having an executive team that includes roles for project management, technical coordination and design, education and training, publicity management and budget control.
- 7) There was a high level of consistency in the 'burning questions' that were raised in the seminar in the different cities. This shows a common and consistent awareness among the audience. Also, the nature of the questions (see section 2.3) shows that SIDP has been successful in showing that the fundamental technical issues have been solved.



4.2 Next steps

Next steps should support the following objectives:

- Finalising outstanding activities
- Further communication and promotional activities
- Ongoing “ownership of maintenance” of specific SIDP outputs
- Maintaining the momentum generated by SIDP
- Actively address ‘burning issues’

The first three are a continuation of SIDP activities and require little or no development of additional materials. The last two require significant additional work and resourcing.

4.2.1 Finalising outstanding activities & further communication and promotional activities

- Darwin workshop and seminar (8 July 2005)
- Participation in SSC 2005 Interoperability Workshop
- Selected, targeted presentations

4.2.2 Ongoing “ownership of maintenance” of specific SIDP outputs

- Ongoing maintenance of SIDP website (ASIBA)
- Maintenance and enhancement of notional architecture (OGCA)

4.2.3 Maintaining SIDP momentum, and actively addressing ‘burning issues’

- Further build upon successful SIDP formula
- Develop demonstrators that address ‘burning issues’, both technical (e.g. security, digital rights management, catalogues) and institutional (e.g. privacy, data quality, IP, costing and charging)
- Coordination of implementation activities to ensure ASDI/Notional Architecture compliance
- Development of enterprise level ‘practice guides’ to help with technical and non-technical implementation aspects.



Appendix A Detailed Survey Results

A.1 Summary of responses

Of the 520 attendees, 302 filled out the post workshop evaluation form, a response of almost 60%, making it a highly representative sample. The responses per category are listed below.



1. Can you please indicate which sector/s you are from: (you may tick more than one option)			
		Response Percent	Response Total
Agriculture		2%	6
Asset / Facilities Management		3.3%	10
Civil / Building Construction		2%	6
Culture & the Arts		0%	0
Defence		1.3%	4
Education		2.6%	8
Emergency Services		15.2%	46
Environmental Management / Monitoring		9.3%	28
Federal Government		15.6%	47
Indigenous Lands		1%	3
Insurance		2.3%	7
Land Development		3.6%	11
Land Titling / Administration		4%	12
Law and Order		4.3%	13
Local Government		8.3%	25
Logistics		0.3%	1
Marketing / Market Research		0.3%	1
Medicine		1.7%	5
Resource Inventory / Mining		4.6%	14
Retail		0.3%	1
Social Services		1%	3
Spatial Information Industry		22.2%	67
State/Territory Government		23.5%	71
Telecommunications		0.7%	2
Tourism		0.7%	2
Transport		5.3%	16
Urban Development / Planning		4%	12
Utilities		6.6%	20
Other (please specify)		14.6%	44



Total Respondents	302
(skipped this question)	2

2. After having attended the workshop how would you rate your knowledge of SIDP and interoperable web services?

		Response Percent	Response Total
Very Good		25.8%	78
Good		57.3%	173
Average		14.9%	45
Poor		1.3%	4
Very Poor		0.7%	2
Total Respondents			302
(skipped this question)			2

3. Has the SIDP workshop been worthwhile for you?

		Response Percent	Response Total
Yes		99.3%	300
No		0.7%	2
Total Respondents			302
(skipped this question)			2

4. Interoperable web services will affect your organisation's competitiveness and export positioning....

		Response Percent	Response Total
Positively		58.4%	174
Neutral		22.8%	68
Negatively		0.3%	1
Not applicable		18.5%	55
Total Respondents			298
(skipped this question)			6



5. Which sessions did you attend?			
		Response Percent	Response Total
Session 1: Executive Briefing		17.1%	52
Session 2: Seminar		12.5%	38
Both		70.4%	214
Total Respondents			304
(skipped this question)			0

6. <i>Executive Briefing (if attended)</i>			
6. To what extent did the Executive Briefing meet your expectations regarding what you had hoped to achieve by attending?			
		Response Percent	Response Total
Well above expectations		10.8%	30
Above expectations		32.5%	90
Met expectations		52%	144
Below expectations		1.4%	4
Well below expectations		0%	0
Not applicable		3.2%	9
Total Respondents			277
(skipped this question)			27

7. Please rate the relevance of the <i>scenarios</i> that were demonstrated:
--



		Response Percent	Response Total
Very relevant		27.4%	77
Relevant		46.3%	130
Of some relevance		21.4%	60
Of limited relevance		2.8%	8
Not relevant		0%	0
Not applicable		2.1%	6
Total Respondents			281
(skipped this question)			23

8. Please rate the relevance of the *technology* that was demonstrated:

		Response Percent	Response Total
Very relevant		37.8%	107
Relevant		53%	150
Of some relevance		7.4%	21
Of limited relevance		0.7%	2
Not relevant		0%	0
Not applicable		1.1%	3
Total Respondents			283
(skipped this question)			21

9. Seminar (if attended)

9. To what extent did the Seminar meet your expectations regarding what you had hoped to achieve by attending?



		Response Percent	Response Total
Well above expectations		9.8%	27
Above expectations		34.9%	96
Met expectations		51.6%	142
Below expectations		2.5%	7
Well below expectations		0.4%	1
Not applicable		0.7%	2
Total Respondents			275
(skipped this question)			29

10. Please rate the relevance/usefulness of the Seminar content:			
		Response Percent	Response Total
Very relevant		28.6%	79
Relevant		58.3%	161
Of some relevance		12%	33
Of limited relevance		0.7%	2
Not relevant		0%	0
Not applicable		0.4%	1
Total Respondents			276
(skipped this question)			28

11. Please rate the relevance/usefulness of the training kit for your purposes:



		Response Percent	Response Total
Very relevant		33.3%	91
Relevant		42.9%	117
Of some relevance		14.3%	39
Of limited relevance		1.8%	5
Not relevant		0%	0
Not applicable		7.7%	21
Total Respondents			273
(skipped this question)			31

12. Which state are you based in?			
		Response Percent	Response Total
ACT		11.2%	32
NSW		16.1%	46
NT		0%	0
Qld		16.5%	47
SA		16.8%	48
Tas		10.5%	30
Vic		14.4%	41
WA		14%	40
Overseas		0.4%	1
Total Respondents			285
(skipped this question)			19

13. Other Comments?	
Total Respondents	74
(skipped this question)	230



A.2 Other Comments

The last survey question asked if respondents had any further comments. There were 72 out of 304 valid responses to this question. These are listed in Table 5.

■ **Table 5 Overview of comment made by survey respondents**

Nr	State	"Other Comments"
1	ACT	Discoverability and fitness of use of data and services were not properly addressed.
2	ACT	This was my first GIS presentation - thanks for the intro to this subject.
3	ACT	Seminar wasn't as technical as i was expecting. Overview of WMS & WFS was useful, however i already had a good understanding of these basics of interoperability and was hoping for more in-depth discussion of technology and business cases. But was good to solidify some of the things i was already aware of.
4	ACT	Well done. Where to next?
5	ACT	Very well run. Influencing politicians for funding to accelerate process?
6	ACT	Please thank all involved - very well done for such 'techy' subject matters.
7	ACT	I regret that i did not attend the Exec briefing. Perhaps i misread the promo info but i was under the impression that i would get all i needed from the second session only. As others have probably missed this it may be worth staging a bridged version in a few months through SSI.
8	ACT	Need more focus on catalog and data discovery (incl. standards)
9	ACT	As a participant in the SIDP project the road show demonstrated what we have been promoting for some time but with added development based on total industry involvement.
10	NSW	Worthwhile.
11	NSW	Extra one hour to hear limitations of GIS users at the coal face - to focus future points to address in the quest for efficient data use.
12	NSW	Had a good understanding of technical details before attending.
13	NSW	Thank you very much.
14	NSW	A worthwhile half day workshop.
15	NSW	Introductions too long - better to get straight into scenarios etc.
16	NSW	Excellent Exec Briefing. Can see the time and effort put in. Keep up the good work.
17	NSW	ABC catalyst might be interested in covering this? Well done on a successful technology outreach event.
18	Qld	Very professional presentation of a complex subject. Excellent material with substance. Not just a showcase - a real kick starter Now we need to focus on the complex institutional arrangements and build momentum for change.
19	Qld	Good show.
20	Qld	Well done, great resources.
21	Qld	Top quality presentation and notes. Great demos. Well facilitated. Will you publish the findings/answers to questions posed during the national roadshow? The Brisbane list of QAA was very useful so im sure the others are too.
22	Qld	Well done - at last a practical, relevant proof of concept.
23	Qld	Excellent. Good cross section of expertise (eg. public/private)
24	Qld	Congratulations - Excellent work on the presentations, documentation and for bringing GIS together and moving forward. Very positive for the industry.
25	Qld	Excellent take home kit.



Nr	State	“Other Comments”
26	Qld	Excellent workshop - thank you.
27	Qld	Very professional, very valuable.
28	Qld	Re Exec. Briefing: The CRC demo was quite poor and out of context with the earlier 2 examples - essentially pushing a specific app - it shouldn't be part of an 'Executive' briefing, other than this excellent work.
29	Qld	Timely operation and distribution of data not necessarily critical, however demonstrating the use of several different data types across environments could be useful in enviro management.
30	Qld	Very well presented seminar - Excellent.
31	Qld	Process/Model would seem to be viable at a national level where coarse level spatial datasets are available. Eg. from Geoscience Australia & BOM. But problems arise at state and local levels where 'culture of sharing' breaks down & organisations may not co-operate (or even be obstructive/hostile) - issues of costs, data integrity etc also kick in. Eg. are local councils going to share their data with other organisations - so changing to a 'culture of sharing for the common good' is crucial for more detailed/large scale analysis. Sharing within organisations is already a problem - sharing outside organisations involves a bigger step, mainly political. Dr. Richard Whitlow Trimap Pty Ltd trimap@bizyweb.com.au
32	SA	Good, reinforced that interoperability capability is real, robust, proven. It has been in use/published for 2+ years. What SIDP showed is not new, but it did increase awareness to non-gis people.
33	SA	Good - Understandable by someone with minimum exposure to GIS/Mapping Skills.
34	SA	I intend to develop a knowledge based expert system for decision support - to manage resources. I think spatial interoperability will help in making informed decisions by use of various relevant data from many stakeholders. I imagine a dream intelligent DSS which will pull out the relevant (up to date) information to make decisions faster, efficient and in a transparent way. University of Adelaide tmurugan@gmail.com
35	SA	Overall and excellent event.
36	SA	Im concerned that the Bureau of Meteorology Service was not accurately reflected in the displays portrayed. The service operates at many more levels than that portrayed as a simple display of data. We supply information and add knowledge - which was not presented. I also accept that this was a demonstration project - and does not reflect mature solutions.
37	SA	Assumed a higher IT understanding than i have. [...]
38	SA	Well done.
39	SA	Felt it missed the point that these services can be used across industries.
40	SA	Excellent to see that the profile of spatial information is being raised and discussed with a business focus. Hopefully these types of forums will allow a structured growth of spatial information.
41	SA	Technological considerations from module 3 was difficult to comprehend - but the business considerations was relevant and coherent.
42	SA	Demonstrate and example of the metadata.
43	Tas	Executive briefing rather drawn out and seminar felt rushed. What is going to happen next to implement this system?
44	Tas	List of speakers and contact details, summary and conclusions made available (online)? Next workshop? Updates?
45	Tas	Suggestion on 'how current is data?' : Add the 'date of data currency' and 'when the next update is' along the layer - just like when books or references are being referred to. Also, might be good to include (data) info in the legend layer as well to reduce risk involved



Nr	State	“Other Comments”
		when untrained user reading the maps.
46	Tas	Chat Fest Inc. Ashley Maher would be more relevant at the end of exec, briefing.
47	Tas	Well done, given your first demo. To me the issues rely on the institutions who are the custodians.
48	Tas	Workshop was a good starting point however further forums in this area need to be held to facilitate the adoption of standards and data sharing in Tasmania. A regional sub-group of ASIBA forming was mentioned but this excludes the major stakeholders such as governments, so this needs to be addressed also. Government hold most of the datasets.
49	Tas	Is the facilitator related to John Cusack? Was the guy from SeeGrid part of the presentation? If so, perhaps he should have been on one of the front tables and introduced accordingly. If not, then he shouldn't have contributed as much as he did.
50	Tas	Discovery tools - one only? [...]
51	Tas	Many thanks.
52	Tas	Demos in first session were great but i think they should be shown again during the second session - people needed to see a practical example of the process.
53	Tas	Demonstration/scenarios were very professionally presented as was facilitation. Well done.
54	Tas	Screen needs to be higher for audience to see.
55	Vic	A good opportunity to [...] and understand the strategic focus for Australia. A strong focus on asset management and local community engagement from a local govt perspective thanks, Bill. Bill Keon City of Melbourne Manager of Asset Services and Municipal Emergency resource Officer 9658 9535 Billkeo@melbourne.vic.gov.au
56	Vic	Excellent presentation and content of training materials
57	Vic	Venue not very suitable for format - seating layout.
58	Vic	Scenarios were a little inaccurate, not reflecting emergency management arrangement in this state ie. example of users. [...] [...]
59	Vic	Room layout poor, unsuitable shape for purpose.
60	Vic	I hope you are noting the questions asked and are going to create a FAQ list for users. It could be worthwhile creating some actual 'real time' demos where users can pull up data live from various servers and then analyse it.
61	Vic	I didn't enjoy the 'business' sessions where the audience was allowed to discuss/whine about costs and licensing. SIDP was the topic.
62	Vic	Explanations of technologies (web server) and how they work was good. Semantic/Legal/Business interoperability needed more mention and explanation (ie. these are core probs rather than tech) during executive briefing. Poor location and acoustics. Conference consultant seemed to 'disrupt' the session rather than facilitate.
63	Vic	Very interesting. I knew nothing about this subject and now i know quite a lot and have material to learn further.
64	Vic	Thanks for your info. May i give other info for future workshops: Xiaogang Chen University of Melbourne x.chen4@pgrad.unimelb.edu.au or xgchenzoof@yahoo.com.au How to assess your services from decision makers and end users or affected people. Sometimes people involved may make inappropriate judgement at end stage or internet breakdown. Limitation of updated data and different map information may affect accurate prediction and realistic judgement, ie. how to deal with it.
65	Vic	[...] I would be interested in being involved in the future as i believe we have something to offer. Mark Karklins Virtual Map Mark@street-directory.com.au 04 25748899 03 98661102
66	Vic	Discussion of file formats would have been helpful - ie. is it ok to have shapefiles - or what if i only have a csv file, are there scripts etc to convert to open source formats.



Nr	State	“Other Comments”
67	Vic	Data availability is the key.
68	WA	Time well spent in obtaining information about SIDP. A very useful presentation and workshop.
69	WA	I found it well worthwhile.
70	WA	Very good to see.
71	Unknown	Room layout could have been better - hard to see over tall heads in front.
72	Unknown	I thought the third presentation of the Exec Summary muddled the message that the technology is available now.



Appendix B ‘Burning Questions’ raised in seminars

B.1 Hobart

- Where are indexes of available services and access levels?
- How do you get data into a GIS?
- How to publish?
- Security – who can see what and who cannot?
- Where do we find out about standards & who coordinates them?
- How to ensure data consistency?
- How software dependent?
- How current is the data?
- Can I use WMS and WFS at the same time?
- If I send data on WFS can I restrict access?
- How to access paid services?

B.2 Melbourne

- Where do I find services?
- Show how to add new services?
- How do we get standards services?
- How do you know how current data is?
- Who sets interoperability standards?
- How do I have confidence that services are available?
- How do we publish our own data without starting from scratch?
- From demonstrator to real?
- How does charging work?
- How to apply security?
- How much do you need to spend?
- How can you mitigate your existing data?

B.3 Sydney

- Rather than pre-set spatial queries (as demonstrated) how do we find/get access to spatial information on the fly – Cataloguing system; commercial arrangements?
- How do we select areas of interest and find data for that area?
- What sort of client software is available?



- Where do I find WMS & WFS services to add to local data?
- What are the issues of liability of data providers and how can they be overcome?
- How does a data provider publish stuff?
- How do changes get propagated?
- How do I supply security to sensitive data?
- Copyright and attribution issues – how to manage?
- Standards re availability, quality and currency of service provided?
- How do you handle conflicting projections (or no projections)?
- What tools are available to support my decision-making processes?
- How much does all this cost?

B.4 Canberra

- How do you pull these services together (from various providers)?
- Contingency planning for when services are not available (risk management)?
- What services are available now, or are likely to be available?
- How do I know which services exist and which ones to use?
- Did we come across any data standards issues and how did we deal with them?
- How does risk get managed – trusted users?
- Ownership of data, access & willingness to share, cost, custodianship?
- Liability of incorrect data use?
- Linkage with spatial and non-spatial data layers?
- Use of wizards for non-technical users (user-centered design)?
- Which standards to use?
- How far has this gone in defining communities and schemas?
- Testing of services in different clients?

B.5 Brisbane

- What's next
- What effort was required in the project?
- Have we got the data to do this?
- What happens to the spatial industry if we don't do this?
- How do we get money for this?
- How do we manage risk of using other data of uncertain quality?
- How do we expose organisational assets while ensuring security?



- What degree of customisation is required?
- Where are the catalogues saying where web services are?
- Reliability of messaging (email) to get web links out?
- Liability issues of published data?
- Overall issues of security (who can use what)?
- Chances of collaborative data quality maintenance?
- How do we find and use other people's models and algorithms?
- Collaboration vs keeping business advantage?

B.6 Adelaide

- How do I present standards-based information to others (both metadata and data)?
- What are the business issues for using interoperability?
- What's next - should the project be doing something?
- Privacy !!
- Disclaimers, data reliability - legal issues....
- Public access to data in an emergency situation - how do you do that?
- Where will this work (ie CFA shed on KI)?
- How future proof are the standards?
- How to providers get paid for information and how do users get charged?
- What impact on data providers - what service level agreements?
- What's this cost?
- What impact of incorrect modeling on business?
- What provision for contingency planning and risk management?

B.7 Perth

- What is the business risk of exposing data?
- What about copyright?
- How much?
- What about mobile solutions?
- What about government pricing of data?
- Business continuity and risk?
- Is this available now?
- Vulnerability of network in emergency management situations?
- How do I combine different web map services (vector and raster)?



- What about remote area connectivity and bandwidth?
- How do I charge for and pay for data?
- How do I find, query, and interpret data?
- What standards are available for data quality?
- How do I know data is authoritative?
- How do I know data is up to date?
- How do I integrate into non-spatial systems?
- What about cultural issues around change management?
- What about public liability?
- How do I know that data has changed in a web map system?